

NAME: _____

PERIOD: _____

DATE: _____

Interpreting a Chi-Square Result

AP Statistics · Unit 3 · Topic 3.15 · B: 2027-01-04 · E: 2027-02-08

[STAR] TODAY'S PROBLEM

Suppose a city library takes an SRS of 240 of its 6,000 active adult cardholders. It records each person's chosen notification method and whether the person returned an item late in the previous six months.

Nia: "Email/Late has the largest raw gap, 13.75; I think that cell drives the statistic and the result shows email prevents late returns."

Omar: "Expected counts differ, so raw gaps may not settle cell influence. I would calculate contributions and examine how notification methods were obtained before interpreting."

Late returns

Method	Late	Not late	Total
Email	15	85	100
Text	24	56	80
None	30	30	60
Total	69	171	240

FIRST TAKE — before the video (5 min)

Which account is better supported so far? Name one calculation and one design fact needed to decide.

[BOOK] SCALE EACH GAP, THEN BOUND THE CONCLUSION (keep on the page)

For a chi-square test of independence, H_0 says the two categorical variables are not associated in the population; H_a says they are associated. Calculate $E = (\text{row total})(\text{column total})/(\text{table total})$ and $\chi^2 = \sum (O - E)^2/E$ with $df = (r - 1)(c - 1)$. Every expected count should be greater than 5, and the p-value is the upper-tail probability assuming H_0 is true. A cell's contribution scales its squared gap by E , so the largest raw gap need not contribute most. Random sampling supports population scope; without random assignment, association does not establish causation.

Finish after the video:

DOK 1 (a) Give the row totals, column totals, table total, and degrees of freedom.

DOK 2 (b) Verify random sampling, 10 percent, and expected counts. Calculate all expected counts, cell contributions, χ^2 , and the upper-tail p-value.

DOK 3 (c) Adjudicate the accounts. Compare the Email/Late and None/Late raw gaps and contributions, use the p-value for a population conclusion, bound causation, and state the two consequences of the rejected account.

Frame: For the two cells, the raw gaps are _____ and _____, while the contributions are _____ and _____; therefore _____.

Frame: The result supports _____ for _____; causation is _____ because _____.

EXIT — turn this in

One thing the video changed about my first take: _____